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Review

Formulation, structure, and applications of therapeutic and amino acid-based deep eutectic solvents: An overview

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ABSTRACT

In the design of greener chemicals, deep eutectic solvents (DESs) are considered as one of the most versatile alternative media with widespread applications. DESs have the advantages of being nonflammable with negligible vapor pressure compared to the traditional solvents. They share many characteristics of ionic liquids, but DESs are cheaper to formulate, typically nontoxic, highly pure, recyclable, biodegradable, and are suitable for use with biological systems. In this article, the two emerging and unconventional types of DESs, therapeutic DES (THEDES) and amino acid-based DES (AADES) are reviewed. Based on the published literature, choline chloride and menthol are the two most common constituents to formulate THEDESs, while proline is the most common component to formulate AADESs. THEDESs have primarily been developed to enhance the efficacy of drugs, whereas AADESs have been explored for pharmaceuticals as well as for the extraction of phytochemicals, radioactive iodine capture, oil separation from ores, and to synthesize bioactive compounds. Since detailed knowledge of their structures, nonbonding interactions, formulations, and applications are yet to be explored, the aim of this review is to compile these details and provide a solid background for the future research.

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1. Introduction

A homogeneous mixture which obtains a melting point lower than their individual components, typically using noncovalent interactions, at a certain molar ratio, is called eutectic system [1]. In 1884, Guthrie first named this system as eutēxia (a Greek word which means well-melting) or eutectics. Since eutectics have been used for a long period, it has been referred to with different names in various areas of research and industries. They are called eutectic mixtures in pharmaceuticals [2], phase change materials (PCM) in the field of thermal energy storage devices or building materials [3–5], eutectic alloys for metallurgy [1], deep eutectic solvents (DES) [6] or low transition temperature mixtures (LTTM) [7] in green chemistry.

DESs are typically comprised of a hydrogen bond acceptor (HBA) and a hydrogen bond donor (HBD). Strong nonbonding interactions, such as dipole, hydrogen bond, alkyl-alkyl interactions, halogen bonds, and Van der Waals forces, exist among the HBA and HBD molecules, which lead to a significant depression of the melting point and inhibit the crystallization process [6]. Smith et al. (2014) described DESs by a general formula, Cat^+X^-zY [8], where Cat^+ refers to the cation of different organic salts (typically choline cation), and X^- refers to the halide anions (typically Cl^-) of the salt or Lewis base, Y refers to a Lewis or Brønsted acid or HBD (e.g., urea), and z refers to the number of HBD molecules. The DES classification system could be extended by including the emerging and unconventional types of DES in the manner shown in Table 1.

Smith et al. [8] considered organic salts as the major hydrogen bond acceptors (HBAs) in their classification, and among them, Type 3 has been studied most. DESs belong to Types 5–7 are comparatively new. However, two emerging types of DESs (e.g., THEDES and AADES) will be discussed in this review. They could be considered as Type 3 or 5, but their specific and widespread applications demand to make separate categories for them. This will also help to track their evolving applications.

Bioavailability of drugs is the key factor for successful drug delivery [9]. Studies showed that about 90% of the lead compounds and 40% of market drugs are poorly soluble in water [9]. Besides, over 40% of new drug candidates require formulation help to improve their solubility and dissolution [10]. Since a huge percentage of drugs have less bioavailability than required [11], a few methods (e.g., lyophilization, amorphous solid dispersion, particle size reduction, complexation, and salt formation) have been using to enhance the bioavailability of such problematic drug molecules [10,12]. Solubility enhancement and changing release kinetics could be the two most promising ways to fix these problems [13]. Optimization of the bioavailability keeping pharmacokinetic variability at minimum levels must be taken under consideration during any new formulation approach [14].

Based on key factors, e.g. solubility, dissolution, and absorption, oral drugs are categorized into four groups. This classification is called 'biopharmaceutical classification system (BCS)', which has been considered as a standard classification system since 1995 [15]. It guides formulation scientists to determine the suitable dosage form and the bioequivalence tests [16]. Table 2 shows BCS classification system with their major properties [11]. Except for some common drug molecules, most of the examples provided in the table have already been studied as eutectics (Tables 3 and 4).

Drugs belonging to BCS Classes I-III are mainly studied in the form of THEDES to observe their efficiency to overcome the problems associated with them [20–22], but the drugs belonging to Class IV have not been explored as eutectics.

Therapeutic DESs require at least one active pharmaceutical ingredient (API) for their formulations. Similarly, for the formulation of AADES, at least one amino acid is needed. APIs and amino acids can act as either the HBAs or HBDs in a DES structure. THEDESs have been formulated to apply in pharmaceuticals or biological fields [18,20–22]. AADES has not only been used for the enhancement of the solubility of drugs and proteins [9,26] but, also used in several other fields, such as improvement of biosynthesis [27], radioactive I_2 capture [28], extraction of phytochemicals [29], oil separation from ore [30], and enzymatic hydrolysis of lignocellulose [31]. Although THEDESs and AADESs are becoming popular, a detailed review article is not available which could help researchers exploring updated information of these fields. Two recent review articles [32,33] have been published focusing on the applications of some API-based DESs for the drug delivery, but they didn't discuss how the THEDESs should be classified, what are the differences of THDES compared to AADES, importance of the study on nanostructures, water content measurements to improve the efficacy, efficient formulations of the DESs considering the green chemistry principles, and other possible applications. Therefore, the aim of this review is to provide a detailed information on the formulations, structural features, and the representative applications of the THEDESs and AADESs reported to date to provide an overview for the future researchers. Besides, our review clarifies that which THEDES are room temperature THEDES (RT-THEDES) and which are not. This classification will help researchers to pick up one of the THEDESs depending on their expected applications, such as green chemistry researchers will try to keep any of their developed process as energy efficient as possible and will try to achieve 100% atom economy. In this review, we have divided the AADES as a separate category and explained why this kind of DES should be a separate domain.

Table 1
Classification of the deep eutectic solvents.

Types	General formulas	Examples
1.	Cat^+X^-: Metal halides	1:2 ChCl: $FeCl_3$
2.	Cat^+X^-: Metal Halides hydrate	1:2 ChCl: $CrCl_3.6H_2O$
3.	Cat^+X^-: HBD	1:2 ChCl: H_2NCONH_2
4.	Metal Chloride: HBD	1:3 $ZnCl_2$: CH_3CONH_2
5.	Nonsalt HBA: HBD	1:1 Citric Acid: Sucrose
6	API* as HBA or HBD (THEDES)	1:1 ChCl: Phenylacetic acid, 3:1 DL-Menthol: Ibuprofen
7	Amino Acid as HBA or HBD (AADES)	1:1 Betaine: L-Histidine, 2:5 L-proline: Glycerol

* API = Active Pharmaceutical Ingredients.

2. Therapeutic deep eutectic solvents (THEDES)

In a therapeutic deep eutectic solvent (THEDES), one of the components must be an (API). Different studies proved that when an API is administered to a biological system in the form of eutectic, their efficiency of membrane transport, intestinal absorption, solubility, and controlled release are significantly improved [20–22]. The most common HBAs of the THEDESs are choline chloride (ChCl) and menthol, which can combine with a large variety of APIs, such as acetylsalicylic acid, ibuprofen, and phenylacetic acid [21,24]. Eutectic mixtures have been used for

Table 2
Biopharmaceutical classification of drugs based on their bioavailability.

Class	Properties	Examples	References
BCS Class I	Behaves like oral solution, high solubility, high permeability, rapid bioavailability	Propranolol, paracetamol, salicylic acid, lidocaine, DL-methionine.	[11,17]
BCS Class II	Low solubility, high permeability, variable bioavailability, need to enhance dissolution rate, suitable for controlled release	Ibuprofen, itraconazole, coenzyme Q10, and Azithromycin	[17–19]
BCS Class III	High solubility, low permeability, low bioavailability, problematic for control release	Ranitidine, tetracycline, acyclovir, Aspirin, peptides, nicotinamide.	[11,17,18]
BCS Class IV	Low solubility, low permeability, poor and variable bioavailability, not suitable for oral drug delivery	DL-methionine, Ciprofloxacin, Neomycin	[17,18]

the formulation of drugs, which could have melting points either below or above the ambient temperature (Tables 3 and 4).

2.1. Formulation

THEDESs are typically prepared by the mixing of two components at a certain molar ratio employing thermal treatment methods. The mixture is heated at 40–80 °C in covered glassware under constant stirring (≤ 600 rpm) at atmospheric pressure until a homogeneous liquid forms [20,23,34]. The heating temperature could be higher depending on the melting points of the constituents used for the formulation of a THEDES. It takes 30 min to 3 h to obtain the homogeneous liquid solution. After the formulation, it should be preserved into a desiccator or stored at −20 °C in a freezer, as the chemicals (e.g., ChCl and menthol) used for the formulations of THEDESs are highly hygroscopic. Different formulations of the THEDESs are listed in Tables 3 and 4. It is observed that ChCl, menthol, urea, and glycerol are the most common constituents. Sometimes the same compound c act as HBA in one THEDES that can act as HBD in another THEDES. For instance, API could act as HBA when their salt form is used for the formulation, but same API could act as HBD when their pure form is used.

2.2. Structural insights/features

Choline chloride (ChCl) and menthol are the two most common constituents to formulate THEDESs. Therefore, structural features of the ChCl and menthol-based THEDESs are discussed in this review. The first category is the ChCl-based THEDES. The nonbonding interactions among the ChCl and HBD molecules play a major role for the formation of THEDES. The types of nonbonding interactions include anion-pi, cation-pi, hydrogen bonds, halogen bonds, and alkyl-alkyl interactions [23]. Among all the interactions, hydrogen bonding between the chloride anion of ChCl and the hydrogen of hydroxyl or carbonyl of the HBDs are the main forces. The major interaction and the orientations of the

Table 3
Therapeutic deep eutectic solvents which have eutectic points above ambient temperature.

HBA	T_m (°C)	HBD	T_m (°C)	molar ratio	T_e or T_g* (°C)	References
Coenzyme Q10	50–52	L-menthol	39.0	6:4 (%w/w)	37.0	[35]
		Lauric acid	43.8	7:3 (%w/w)	37.9	[36]
Ibuprofen	75–77			3:7	35.2	[37]
		Thymol	49.6	2:3 (%w/w)	32.0	[21]
Lidocaine	68.0	DL-camphor	174–179	1:1	33.0	[38]
		Tetracaine	148 [39]	1:1	30.0	
Paeonol	52.5	DL-menthol	39.0	4:6	≤32.0	[40]
Lidocaine	68.0	Stearic acid	69.3	1:1	43.0	[41]
Testosterone	151.0	DL-menthol	39.0	1:4	39.9	[42]
Ibuprofen	75–77			1:3	−50.1*	[20]
ChCl	305.0	Salicylic acid	159.0	1:1	68.0	[43]
		Thymol	49.6	1:2	67.0	
		Paracetamol	168.0	1:1	41.0	
Tetracycline	170–175	Glycerol	18.1	1:2	62.0	
Ticlopidine	189.0			1:2	67.0	
Borneol	202.0	DL-menthol	39.0	2.5: 7.5	N/A	[22]
Myristic Acid		S-limonene	−90.0	1:1	47.7	[44]
				1:2	47.5	
				2:1	51.6	
Ibuprofen	75–77			8:1	AT	
Sulfathiazole	202.5	Urea	133.0	5.2:4.8 (%w/w)	118.0	[45]

**NA = Not Available, T_m = melting point, T_e = eutectic point, T_g = glass transition temperature, AT = ambient temperature, and the T_m of the HBAs and HBDs are also listed from PubChem [46].

Table 4
Therapeutic deep eutectic solvents which have eutectic points below the ambient temperature.

HBA	T_m (°C)	HBD	T_m (°C)	Molar ratio	T_e or T_g* (°C)	References
Propranolol	96.0	Decanoic acid	31.9	3.5: 6.5 (% w/w)	15.4	[47]
		Lauric acid	43.8	3:7 (% w/w)	15.9	
Ibuprofen	75–77	Methyl nicotinate	38–43	1:1 (% w/w)	AT	[48]
		DL-menthol	39.0	2.5: 7.5 (% w/w)	13.0	[21]
		L-menthol	39.0	3:7 (% w/w)	19.0	
		1,8-Cineole	1.5	2:3 (% w/w)	−13.0	
		Lidocaine	68.0	1:1	−27*	[49,50]
Itraconazole	166.2	Phenol	40.9	2.4: 7.6 3.1: 6.9	<0	[51]
Lidocaine	68.0	Lauric acid	43.8	N/A	6.0	[52]
		Prilocaine	37.4	1:1	18.0	[53,54]
ChCl	305.0	Resorcinol	109.8	1:1	6.0	[43]
		Aspirin	135.0	1:2	−67.0*	
Ranitidine	69–70	Glycerol	18.1	2:1	−87.0*	
Phenformin	176.5			2:1	−69.0*	
Ranitidine	69–70	Urea	132.7	2:1	−31.0*	
Adiphenine	113.5			2:1	−45.0*	
		Aspirin	135.0	2:1	−23.0*	
Ranitidine	69–70			2:1	−37.0*	
Lidocaine	68.0	Decanoic acid	31.9	1:1	−61.0*	[41]
		Hexanoic acid	−3.0	1:1	−56.0*	
		Linoleic acid	−6.9	1:1	−71.0*	
		Oleic acid	16.3	1:1	−47.0*	
		Phenyl salicylate	76.7	3:2	18.2	[55]
ChCl	305.0	Aspirin	135.0	1:1	AT	[24]
		Phenylacetic acid	76.7	1:1		
DL-menthol	39.0	Aspirin	135.0	3:1		
		Phenylacetic acid	76.7	2:1 3:1		
		Decanoic acid	31.9	1:4	17.0	[56]
Decanoic acid	31.9	Limonene	−90.0	1:1	14.3	[44]
				1:2	7.8	
				2:1	20.9	
DL-menthol	39.0			1:1	AT	
				1:2		
				2:1		
ChCl	305.0	Ascorbic acid	190–192	2:1	AT	[34]
Acrylic acid	13	Lidocaine-HCl	80–82	3:1	−78.4*	[57]
Methacrylic acid	16				6.8	

**NA = Not Available, T_m = melting point, T_e = eutectic point, T_g = glass transition temperature, AT = ambient temperature, and the T_m of the HBAs and HBDs are also listed from PubChem [46].

components are shown in Fig. 1, where 1:1 ChCl: phenylacetic acid (PAA) is the THEDES. Experimental and computational IR, Raman vibrational spectral analysis, as well as the molecular dynamics simulation, radial distribution function, density functional theory calculations, and solvatochromism are reliable tools to demonstrate the interactions [23].

It is crucial to understand the structural orientations of the constituents of a THEDES to get the reasoning behind its eutectic nature. Noncovalent interactions predominantly determine their thermal stability, molecular vibration, melting point, density, and viscosity. Choline chloride-based THEDESs exhibit a highly interconnected hydrogen-bonding network, wherein the choline cation and -OH functionalities of HBA and the functionalities of the HBD molecules interact strongly with the Cl^- anion [23]. Besides, methyl protons on choline could interact via C-H…π interactions, depending on the HBDs used for the formulation. Fig. 1 shows that chloride anion could also form anion-π interactions with the phenyl group of PAA, and choline cation could form cation-π interaction with the same aromatic-pi electrons of the PAA. One of our recent studies on 1:1 ChCl-Aspirin (ASA) THEDES showed that the nonbonding interactions (e.g., hydrogen bonds and electrostatic interactions) play a major role to form the eutectics and the distances of hydrogen bonds of the stable cluster conformers are ranges from 1.7 to 2.5 Å [23]. Charge delocalization and solvatochromism studies proved that Cl^- forms a charge transfer bridge between the choline cations and the HBD molecules [23].

Another major category of THEDES is menthol-based eutectics. Structural properties and nonbonding interactions of a menthol-based DES could be investigated with the same tools listed earlier. Structures of these THEDESs have not been studied as much, but a few studies suggested that the menthol interacts with the other constituents using its -OH group and the lipid alkyl tail [58,59]. For example, menthol interacts with the ibuprofen molecule by H and O atoms of its OH, where the contributing atoms of the ibuprofen are O of the C═O and H of OH of the —COOH (Fig. 2) for 3:1 DL-menthol:ibuprofen THEDES. Interactions from the lipid alkyl tail could be alkyl-alkyl interaction and unconventional C—H hydrogen bond [59]. FTIR studies on the THEDES formulations reported in Stott et al. (1998) article also proved that the ibuprofen forms THEDES with thymol (Tables 3 and 4) by forming strong hydrogen bonds among their carbonyl and hydroxyl groups [21].

3. Applications of THEDES

Applications of THEDES have not been rigorously studied. These applications can be categorized in the following ways, (i) interactions of the THEDESs with the biological system: skin permeation and intestinal absorption, (ii) physicochemical changes: dissolution and solubility enhancement, (iii) controlled drug release, and (iv) other applications: improvement of anti-cancer drug delivery by the formulation of THEDESs.

3.1. Permeation enhancement of transdermal drug delivery system

Transdermal drug delivery is one of the most popular administration routes for analgesics because it offers several advantages, such as the extension of the duration of activity, minimization of pain, reduction of side effects, and controlled drug release. Since it often faces problems related to permeation through the membrane, eutectic formulations have been used to overcome the problems. Some applications of the eutectics are following:

Adela et al. studied on 1:1 lidocaine (L):prilocaine (P) eutectic mixture, where both the constituents are two local anesthetics [54]. They investigated that the efficiency of their release of L and P on oil-water dispersed phase and their solubilities through the membrane separately and in the form of eutectic. The enhancement of the distribution coefficient of the L and P were observed while applied in the form of THEDES. Besides, the drugs had higher(escaping tendency in liquid phase when they are administered as 1:1 L:P. This mixture was further emulsified in water with a surfactant to overcome the limitations (lower permeability and drug release rate) of the conventional oil-water emulsions of the local anesthetics. he authors also found that the permeability and the rate of drug release of the mixture in aqueous solution increased with the increase of the concentration of the L-P mixture.

Lidocaine is an anesthetic that can either be injected into the body to act as a nerve blocker or applied topically on the surface of a tissue or mucus membrane to act as a local anesthetic [60]. Lidocaine is very soluble in both organic and polar solvents [61] allowing movement through the skin to the binding sites through the blood with no issue. The major problem that belongs to the drug delivery of lidocaine is transdermal permeation because of its high melting point. Wang et al. (2014) improved the efficacy of lidocaine drug delivery as by formulating a eutectic mixture with ibuprofen [50]. They performed the study using a Franz cell and silicone membrane and concluded that the solubility of lidocaine and ibuprofen were enhanced when applied as the THEDES form. Even in solution, they can maintain strong intermolecular hydrogen bonding with each other which allows them simultaneous transportation through membrane with much higher transport rates than their corresponding commercial crystalline salts. This should lead

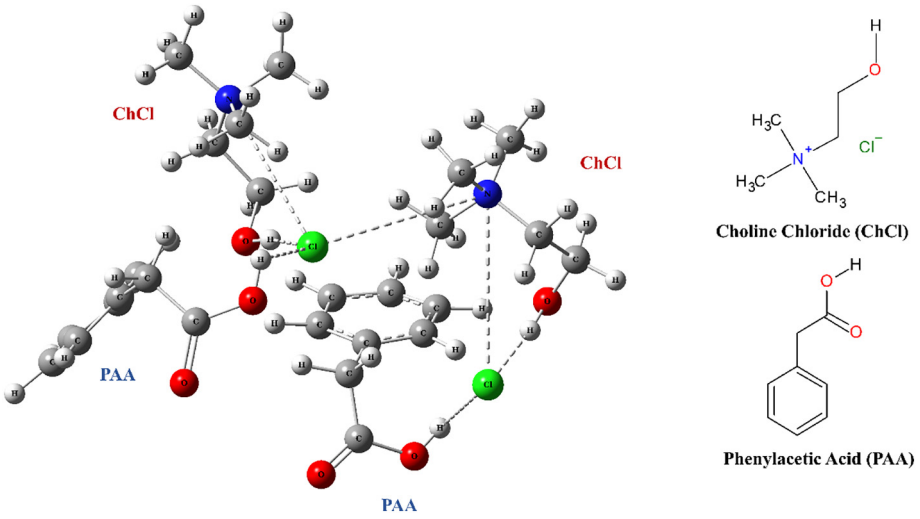


Fig. 1. Nonbonding interactions among the components of choline chloride (ChCl):phenylacetic acid (PAA) (1:1) THEDES and the two-dimensional structures of its components are shown.

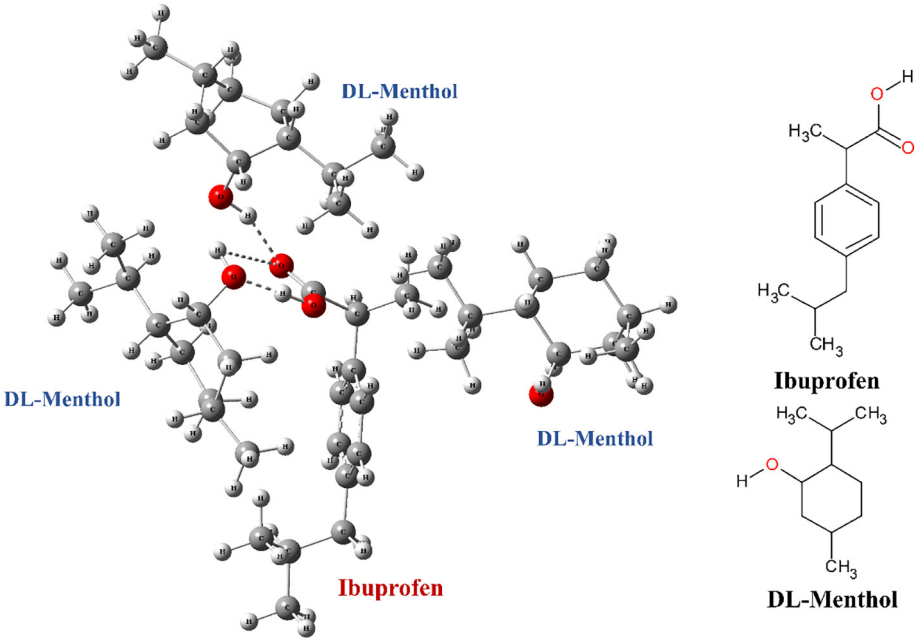


Fig. 2. Major nonbonding interactions in D, L-Menthol:Ibuprofen (3:1) THEDES structure and the two-dimensional structures of its components.

to a range of transdermal treatment options where two strongly interacting APIs can be combined to influence the membrane transport synergistically and provide simultaneous delivery to the blood stream.

Berton et al. (2017) studied the same eutectic mixture using an animal model. They topically applied the mixture to Sprague-Dawley rats by dissolving the mixture (e.g., 1:1 Lidocaine: Ibuprofen) in a vehicle cream [49]. The concentrations of the APIs in blood plasma were monitored over the time as an indication of systemic absorption. Faster and higher systemic absorption was observed for the hydrogen bonded DES than the absorption of the salt forms e.g. Lidocaine-Cl or Lidocaine-Docusate. Interestingly, a differential transdermal absorption was observed between lidocaine and ibuprofen when the THEDES form was applied. It indicated higher interactions among the tissue and the components of the cream, which could facilitate permeating through the skin as a neutral complex and enter systemic circulation.

Stott et al. investigated transdermal delivery of a model drug (e.g., ibuprofen) by forming its eutectic systems with terpenes to observe the enhancement of the drug permeation [21]. The molar ratios and the melting points of the mixtures are listed in Table 3. Ibuprofen permeated across the human epidermal membrane from the THEDESs manifolds than the saturated aqueous solution. For instance, 2:3 (%w/w) ibuprofen:thymol produced a flux of 150 mg/cm^2/h, which was 5.9 times the flux produced from the saturated aqueous solution of thymol in the case of terpene pretreated skin. Moreover, it was 12.7 times higher than the flux from the saturated aqueous solution across non-pretreated skin. FTIR studies indicated that the terpene used to formulate the THEDES forms hydrogen bonds with the ibuprofen, which results in the depression of their melting point, and eventually facilitates the enhancement of the permeation.

Itraconazole,belongs to BCS class II and is an antifungal drug which has high logP value and molecular weight. Low aqueous solubility and high hydrophobicity of the itraconazole could cause irregular absorption and insufficient plasma concentration. To overcome these issues, Park et al. (2012) studied the permeability of the itraconazole by topical administration by forming their eutectic mixture with phenol [51]. They used hairless mouse skin to perform the experiment. The eutectic mixtures of 2.4:7.6 and 3.1:6.9 itraconazole:phenol were applied through the skin using a Franz diffusion cell. The amount of itraconazole permeated through the skin was determined using high-performance liquid (HPLC). They found that despite the high logP value and molecular weight, itraconazole can permeate through skin in the form of THEDES. In this way, itraconazole could be administered topically and could overcome the problems associated with the oral dosage form, such as inadequate plasma concentration and erratic absorption.

Kasting et al. [62] proposed a relationship between transdermal flux and melting point of the permeant based on ideal solution theory. They stated that the ideal solubility of a permeant in the skin lipid could be increased approximately exponentially by decreasing its melting point for any given molecular weight. Results indicated that a reduction in the melting point of a permeant will have a direct effect on its solubility in skin lipids and can increase the transdermal permeation.

Duarte et al. (2017) comparatived experimental permeability of the APIs including ibuprofen, benzoic acid, and phenylacetic acid and their permeability in the form of THEDES [25]. In brief, they measured the permeability using a glass Franz-type diffusion cell and a polyethersulphone (PES-U) membrane. The membrane was placed between the two compartments while the receptor compartment was filled with PBS solution, and the donor compartment was loaded with the same amount of API in the powder form or in the form of THEDES with a few mL of PBS. Time periods for the diffusion was taken for 5 min and hourly from 1 to 8 h at 37 °C. The amount of API diffused was determined using a microplate reader. Permeability of ibuprofen, benzoic acid, and phenylacetic acid have increase about 3, 7.5, and 1.1 times, respectively when applied as THEDES.

3.2. Intestinal absorption enhancement

Shen et al. used 3:1 menthol:borneol THEDES and its microemulsion for the study of daidzein absorption in the rat intestinal membrane using a diffusion chamber system [22]. The microemulsion was composed of ethyl oleate (oil), cremophor RH40 (surfactant), PEG400 (co-surfactant), and water. Both the eutectic mixture and its microemulsion enhanced the intestinal absorption of daidzein in vitro. Bioavailability of the daidzein eutectic and the eutectic-microemulsion were increased by 1.5 and 3.7 times compared to the daidzein suspension. They found that when 3% of the eutectic mixture and microemulsion (containing 3% eutectic mixture) were added to the mucosal surface, the absorption of daidzein was significantly increased, although the mechanism of

absorption enhancement is still unclear. Since daidzein has been used as a traditional medicine, the safety profile should be good [63].

3.3. Control release of drugs

Leija et al. transformed lidocaine hydrochloride into THEDESs by incorporating polymerizable counterparts (e.g., acrylic and methacrylic acid) [57]. These polymer-drug complexes were synthesized by free-radical frontal polymerization through forming hydrogen bonds among the lidocaine-salt and the polymers, where Cl^- works as a bridge. The properties of the eutectics allow frontal polymerization in bulk with full conversion, which was achieved in a one-pot synthesis. It yielded monoliths of polymers loaded with a high concentration of drug. In vitro experiments proved that the drug can be released in a controlled manner triggered by the pH, ionic strength, and solubility of the drug in the dissolution medium. The control release of the drug was maintained by swelling of polymers and the specific interactions among the drug molecules and the polymers. They concluded that the controlled drug release could be achieved not only by the monomers and THEDESs studied but also varying the monomers (as HBD) and crosslinkers. Similar polymer systems were also used for the DES to control drug delivery. Pradeep Kumar et al. [64] used an amphiphilic polymer with DES for tumor-targeting cancer therapy. In this case, DES (formulated using CABAL (3-(4-(4-(bis(2-chloroethyl)amino) phenyl)butanoyloxy)-*N*,*N*,*N*-trimethylpropane-1-aminium chloride) and 1,4-butanediol (BD) in various ratios) influenced the formation of FA-g-β-alanine-co-PCL which eventually can act as a doxorubicin (DOX) carrier. They applied the system against MDA-MB-231 cancer cell line (in vitro) and breast cancer cell model (in vivo). The amide bond of the carrier (DES@FA-g-β-alanine-co-PCL) was degraded by the lysosome in cancer cells, which facilitates the control release of DOX and enhances the anticancer effect. In another study [65], they used proline-lactic acid-based DES to form similar micelle (DES-g-PGA (polyglutamic acid)-FA (folic acid)) to control release of the anticancer drug Paclitaxel (PTX). Their study demonstrated that the DES-based drug carrier are efficient for anticancer drugs, which can minimize the adverse side effects to the normal cells.

In the previous studies, the DESs were used as the carrier of a drug molecule, but Aroso et al. (2015) [20] formed a THEDES (3:1 menthol: ibuprofen) to improve the efficiency of ibuprofen release at the target. The THEDES system was used to design an advanced drug release system employing supercritical fluid technology. To investigate this new approach, they performed a comparative dissolution study of the ibuprofen-powder and ibuprofen-THEDES. They found similar dissolution efficiency for the pure ibuprofen and the ibuprofen in the form of THEDES. The similar efficiency profile indicates that in the solution the nonbonding interactions among the menthol and ibuprofen molecules might be dissociated. Furthermore, they conducted the drug release study in the buffered saline by impregnating ibuprofen and ibuprofen-eutectic in a carrier (starch:poly-ε-caprolactone polymeric blend or SPCL). This highly porous carrier has melting point at 60 °C, which facilitates the application in biological systems. They found a faster release profile of ibuprofen in the form of THEDES, which is mainly governed by diffusion.

Silva et al. performed a similar study [34] using the same carrier (SPCL) and they also got a faster rate of drug release. However, they did not apply the THEDES (2:1 ChCl:ascorbic acid) directly as a drug system, rather they used the THEDES as a solubilizing agent of dexamethasone (DEX). A controlled drug delivery system was developed by supercritical fluid sintering of a SPCL blend doped with THEDES-solubilized DEX. Aliquots were collected at different times and analyzed employing HPLC. This study suggested that the THEDES is not only used for the formulation of a drug but they could also be used as a solubilizing media.

Recently, they formulated some ternary THEDESs, e.g. citric acid:L-arginine: H_2O at molar ratios of 1:1:7, 1:1:6, 1:1:5, and 1:1:4 [66]. They used these THEDESs systems for tuberculosis therapy employing supercritical carbon dioxide technology. Besides the drug release study, cell viability study or in vitro cytotoxicity assay of the THEDES was evaluated against L929 cell line. This study proved that along with the advantages of the improvement of drug release and dissolution profiles, the THEDESs do not compromise the cell viability. Therefore, THEDESs seems safer to use.

3.4. Solubility and dissolution enhancement

Duarte et al. (2017) also studied solubility enhancement of the APIs (e.g., ibuprofen, benzoic acid, and phenylacetic acid) similar to their study on permeability [25]. The solubility measurements were performed with the pure APIs and their THEDES formations. Briefly, an excess of API and THEDES saturated solutions were prepared in PBS solution at room temperature (20 °C) by stirring for 24 h. The determination of the solubilities of the APIs were performed by a microplate reader. They found that the solubilities of the APIs were increased in the form of 3:1 menthol:ibuprofen, 3:1 menthol:benzoic acid, and 2:1 menthol:phenylacetic acid were increased about 12.8, 2.1, and 1.2 times, respectively. Therefore, THEDES can offer enhancement of permeability, dissolution, and solubility, although the mechanism of action has not been fully understood yet.

Win et al. [45] studied the dissolution rate of the sulfathiazole and urea eutectic mixture. This study showed that the dissolution rate of sulfathiazole in the form of THEDES is 700 times higher than its pure form. The formation of the eutectic mixture of 5.2:4.8 (%*w*/w) sulfathiazole: urea (Table 3) enhanced the reduction of the crystal size of the API and made it highly soluble in water. Solid dispersion of the sulfathiazole increased the physiological availability of the drug and showed 12 times faster dissolution rate than its pure form.

3.5. Improvement of anti-cancer drug delivery by the formulation of eutectics

Although limonene has chemotherapeutic properties against many cancers (e.g., breast, lung, prostate, and gastric), it is highly toxic even to the healthy cells [44]. Direct use of it to treat cancer cells is not feasible in terms of cell viability. Therefore, Pereira et al. (2019) studied the potential use of some limonene-based THEDES to treat cancer [44]. They formulated a total of 14 THEDES using limonene, among them only 1:4 ibuprofen:limonene showed successful inhibition of HT29 cell proliferation without compromising the cell viability. H29 cell line was incubated for 24 h to perform the antiproliferation assay by treating the components and the THEDESs. Cytotoxicity assay was assessed with the human Caco-2 cell line. After seven days of cell culture, the cell line was incubated with the THEDESs for 24 h. EC_{50} values showed that the 1:4 ibuprofen:limonene can reduce the proliferation of HT29 cell line without compromising the cell viability. It proved that ibuprofen could have a synergistic effect to the limonene for the anticancer treatment. It also led to the enhancement of anti-inflammatory activities of ibuprofen, which could indirectly help the anticancer therapies.

4. Amino acid-based deep eutectic solvent (AADES)

Amino acid-based deep eutectic solvents (AADES) are another type of emerging and potentially green solvent, which is naturally benign, biodegradable, sustainable, and eco-friendly [67,68]. To formulate AADESs, at least one of the components should be an amino acid. AADESs have been used in various areas of research, such as cosmetics, food, drugs, animal feeds, and polymer production [69,70]. They have shown a lot of promise in several other fields, such as solubility enhancement of drugs and proteins [9,26], biosynthesis [27], radioactive I_2 capture [28], extraction of bioactive compounds [29], separation of oil-carbonate [30], and depolymerization of lignocellulose [31].

4.1. Formulation

Two methods have been primarily used to synthesize AADESs, vacuum evaporation and heating. For the vacuum evaporation method, components are dissolved in water, mixed, and heated. Their dissolution could be enhanced by vortexing or applying ultrasound, and finally the water is evaporated [71]. In the heating method, components of the DESs are mixed at a certain molar ratio and heated in covered glassware under constant stirring until a homogeneous liquid forms [68,71]. Studies showed that most of the AADESs have been prepared using heating. To synthesize an AADES, selected components are mixed at a certain molar ratio (range of 10:1–1:10). Then the mixer is heated at 25–100 °C with constant magnetic string at 400–1200 rpm for 0.5–24 h until a clear and homogeneous solution forms, depending on the nature of the components and their molar ratio [30,72,73]. Synthesized eutectics are dried at 45 °C in a desiccator over P_2O_5 [74]. Recent studies suggested that water content can significantly alter the physicochemical behavior of a DES. Kuddushi et al. (2019) reported that if more than 25% water is added to a DES system, then the added water molecules could take control of the formation of H-bonds and eventually could reduce the efficacy of the solvent systems [75,76]. Savi et al. (2019) experimented that after adding 10% of water to 5:1 Lactic Acid: Glucose, viscosity of the solvent got decreased to 85.9% [77]. Therefore, it is important to determine the water contents of a formulated DES. Since most of the AADES formulations listed in Table 5 are focused on their specific applications, quantifying water contents in the formulations are not focused that much or they might not report them in the respective articles. Dai et al. (2013) determined that most of their prepared AADESs showed close to 0.2 mol (%) water activity values. Besides, Mjalli (2016) and Chen et al. (2020) measured maximum 500 and 800 ppm water contents in their formulated AADESs, respectively.

4.2. Structural features

Formation of a DES depends on the strength of the noncovalent interactions among its constituents, which directly affects the physical and chemical properties, stability, and phase-transition temperature of the solvents [83]. Different techniques have been applied to explore the structure of AADES, such as ^{1}H NMR, FTIR, and Raman spectroscopy [7,23]. Based on the noncovalent interactions, structures of the AADES can be classified into two types. The first is formed by the electrostatic interaction, e.g. quaternary ammonium salt-based AADES or betaine (trimethyl glycine or $(CH_3)_3N^+CH_2CO_2^-$)-based AADES. Since it does not have any halide anion, betaine and the amino acid molecules can directly interact with each other using electrostatic forces. Hydrogen bonds among the functional groups (e.g., -COOH and -NH) could also facilitate formation of the AADES [31,84]. For better understanding, those interactions are shown in the Fig. 3.

The second major type of AADES is formulated by nonionic hydrogen bond acceptors and donors. In this class, proline-based AADES is the most common type. Dai et al. (2015) investigated the formation and the structure of 1:1 proline:malic acid using NOESY (^{1}H–^{1}H-nuclear overhauser enhancements) and FTIR spectroscopy [85]. They revealed that O═C-O-H—N is the major hydrogen bond in the DES system. That means -COOH of the malic acid is donating a hydrogen bond to the N of proline. Their study also demonstrated that when the molar ratio of proline-malic acid increases, the number of hydrogen bond increases. Besides, alcohol group containing HBDs are another major type of HBDs for the AADES formulation. To understand their structure, 2:5 L-proline:glycerol [29] is considered being shown in Fig. 4. Glycerol is an efficient HBD for the formulation of many types of DESs because it offers long range diffusivity. Therefore, it can form a strong hydrogen bonding network in the DES system [86].

Table 5
Different combinations of amino acid-based deep eutectic solvents and their HBAs, HBDs, and molar ratios are also listed.

HBA	T_m (°C)	HBD	T_m (°C)	Molar ratio	T_e or T_g* (°C)	References
Bu_4NBr	103.0	Aspartic acid	270.0	1:9	25.2	[72,78]
				1:10	32.8	
				1:11	31.8	
		Glutamic acid	199.0	1:8	30.6	
				1:9	29.3	
				1:10	28.9	
		Arginine	260.0	1:6	25.1	
				1:7	26.4	
				1:8	26.2	
Betaine	293–301	Lysine	224.5	1:1	AT	[31]
		Arginine	260.0	1:1		
		Histidine	287.0	1:1		
Sucrose	186.0	DL-Proline	208.0	1:2, 3:1 1:4	<−50*	[79]
Lactic acid	16.8			1:1		
Citric acid	153.0			1:1, 1:2		
DL-Malic acid	131.0			1:1		
D-(+) Glucose	146.0			1:1, 3:5		
D-Sorbitol	95.0			1:1		
D-(+) Glucose	146.0	L-Proline	205.0	3:5		
		D-Proline	223.0	3:5		
DL-Malic acid	131.0	L-Serine	246.0	1:1, 2:3		
		β-Alanine	207.0	1:1, 2:3		
Citric acid	153.0			1:1		
L-Cysteine	240.0	Lactic acid	16.8	1:8,	NA	[28,80]
L-Methionine	281.0			1:12,		
L-Tryptophan	290.0			1:16		
Glycine	233.0					
Valine	298.0					
L-Leucine	293.0					
L-Proline	205.0	D-glucose	146.0	1:1	NA	[29]
		Sucrose	186.0	2:1		
		D-sorbitol	95.0	1:2		
		Glycerol	17.8	2:5		
		Citric acid	153.0	1:1		
		Laevulinic acid	33–35	1:2		
		Oxalic acid	189–191	1:1		
		Lactic acid	16.8			
		DL-malonic acid	131.0			
		Malonate	199.0			
		Urea	133.0			
		L-methyl urea	99.0			
		Acetamide	221.2			
Arginine		Oxalic Acid		1:1	AT	[81]
		Tartaric Acid			AT	
		Glutamic Acid			~30.0	
DL-Proline	208.0	Oxalic acid dihydrate	101–102	1:1	AT	[82]

**NA = Not Available, AT = Ambient Temperature, T_m = melting point, T_e = eutectic point, T_g = glass transition temperature, and the T_m of the HBAs and HBDs are also listed from PubChem [46].

5. Applications of AADES

5.1. Biological applications

Amino acid-based DESs show remarkable efficiency to improve bioavailability, synthesis, and conversion of drugs. L-glutamine and glycine (2:5) AADES media can increase the biosynthesis of ethyl (*S*)-4-chloro-3-hydroxybutanoate with the increase of the reductase activity by 1.67-fold compared to the water control media [27]. He et al. synthesized a novel AADES with the combination of one molar choline acetate

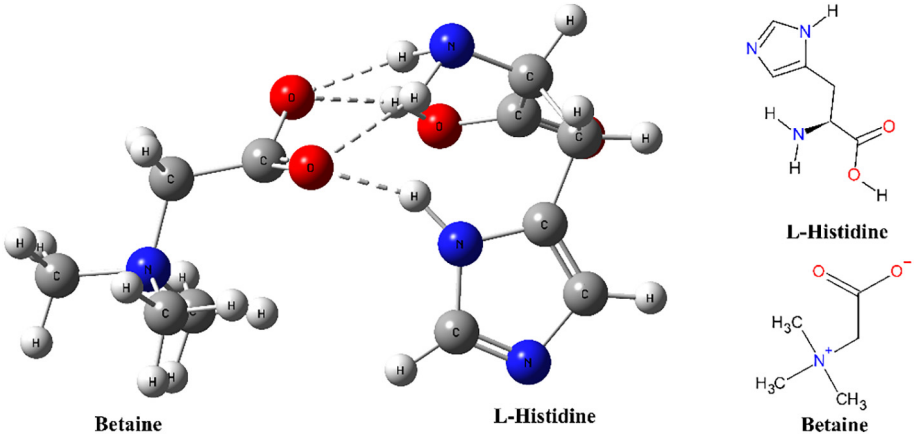


Fig. 3. Major interactions are illustrated in the cluster structure of 1:1 betaine:L-histidine, where betaine is the hydrogen bond acceptor and L-histidine is the hydrogen bond donor molecule.

(ChAc) and one molar lysine to improve the reaction efficiency for whole-cell-catalysis [74]. In their study, asymmetric bioconversion of CFPO (2-chloro-1-(3,4-difluorophenyl)ethenone) to (*S*)-CFPL ((*S*)-2-Chloro-1-(3,4-difluorophenyl)ethanol) occurred in the AADES mediated by recombinant *E. coli* cells. It was used as a model reaction, where the AADES were assessed as an excellent media for their performance in bioreduction. It was observed that the AADES can improve the efficiency of coenzyme regeneration and permeability of cell membrane during the bioreduction process. It can also increase the substrate loading by 3.3-fold compared to the aqueous media.

Stability is a key factors in the protein chemistry to avoid unnecessary aggregation and denaturation. Proteins can be aggregated by storage conditions, incorrect folding during protein expression, perturbation during purification, formulation, freeze-thawing, drying, ultrafiltration, and transportation [87]. Since aggregation of a protein is a big problem, different additives have been used to enhance the stability. L-Arginine is one of the widely used additives to enhance protein refolding by inhibiting aggregation [88,89]. Golovanov et al. showed that L-arginine and L-glutamic acid separately can enhance protein stability, but an equimolar mixture of both synergistically could enhance the solubilities of the proteins studied 4.1–8.7 times their solubilities in the buffer solution [26]. They also interact with the oppositely charged residues of the protein, water molecules, and different ions in the buffer. Thus, it forms an alternative to stabilize proteins. The equimolar mixture could effectively reduce protein-protein interactions. Shukla et al. (2011) modelled the synergistic effect of L-arginine and L-glutamic acid on protein stability by simulating the system using molecular dynamics [90]. Their study revealed that the stability of the protein is related to the increase of the number of L-arginine and L-glutamic acids around the protein. An equimolar mixture of L-arginine and L-glutamic acid could increase the presence of additional molecules on the surface of the protein, increasing its stability.

Arginine-based DES could play an important role as a drug solubilizing agent [81]. In pharmaceuticals, one of the major concerns is the poor solubility of the APIs [9]. Studies showed that a huge number of the lead compounds and market available drugs are poorly water soluble, and the cost of launching a novel drug is extremely high [9,91]. Therefore, it would be a better approach to improve the efficacy and bioavailability of the existing drugs in their formulation stage [92]. Similar to the THEDES, arginine-based DESs have been reported as good drug

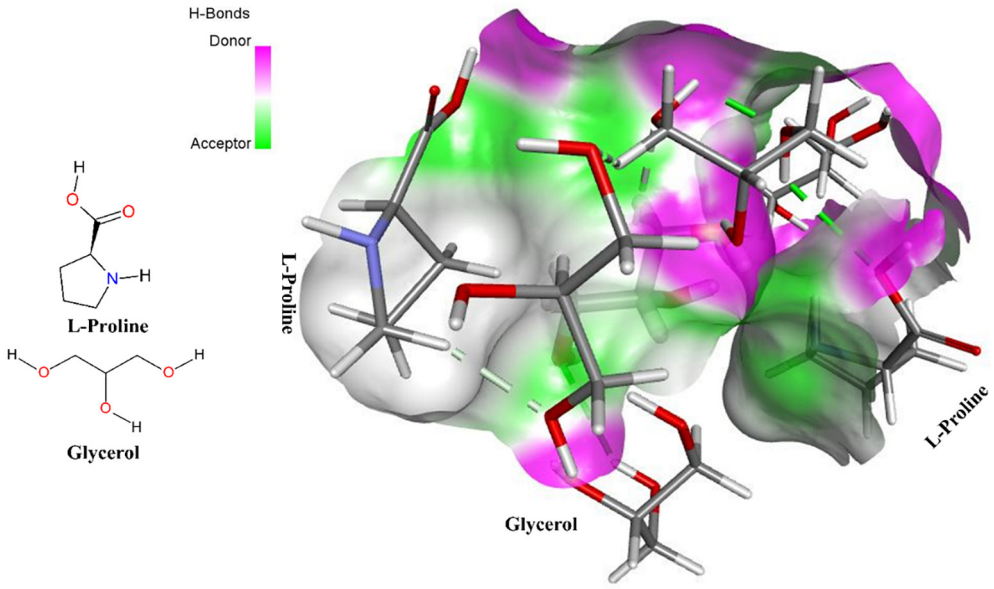


Fig. 4. Glycerol is diffusing in the structure of 2:5 L-proline:glycerol AADES, where the L-proline is the hydrogen bond acceptor and glycerol is the hydrogen bond donor molecule.

solubilizing vehicles [81]. In this study, solubilization of lidocaine was studied in the arginine-based DES including (e.g., 1:1 M ratio of arginine with glutamic acid, oxalic acid, or tartaric acid). Results showed that the interactions among the lidocaine and the constituents of the DESs are responsible for the enhancement of the lidocaine solubility. Molecular dynamics study also showed that the proper selection and tuning of the HBDs and HBAs of a DES could significantly impact on their efficiency to enhance the drug solubility [81].

5.2. Radioactive iodine capture

Nuclear energy is one of the most efficient and clean energies. However, it produces an enormous amount of radioactive nuclear waste, some of which could cause serious damage to the environment even by a small release and it could cause cancer [93,94]. Therefore, the radioactive I_2 requires efficient entrapment to prevent its fatal effect to the environment and human health [95]. Studies recommended that some DESs are efficient to capture the radioactive I_2. For instance, Chen et al. captured radioactive I_2 using AADESs, such as cysteine: lactic acid, methionine:lactic acid, tryptophan:lactic acid, glycine:lactic acid, valine: lactic acid, and leucine: lactic acid, which are formulated at different molar ratios [28]. Their study showed that AADES can efficiently capture the radioactive iodine. Among the AADESs studied, 1:8 cysteine:lactic acid performed best and 1:8 leucine:lactic acid least well as the 1:8 cysteine:lactic acid can form a higher number of halogen bonds with the iodine.

5.3. Oil separation

Different types of oil-solid separation techniques (e.g., solvent extraction, pyrolysis, and microbial enhanced oil recovery) have been using in the fields of unconventional petroleum exploitation, pipeline deoiling, paper deinking, oil-contaminated water, and food industries [96–99]. AADES-based separation is now considered one of most promising green approaches to separate oil and solid materials. Zhang et al. synthesized a novel amino acid ionic liquid-based DES (e.g., TrpBF_4/U) for the extraction of oil-carbonate minerals from the Indonesian asphalt ores [30]. They reported that amino acid ionic liquid-based DES can increase oil-recovery about 11% compared to the oil-recovery technique based on traditional ionic liquid (e.g., [Emim]BF_4). This is because DES can reduce of the interactions between the oil and mineral surfaces.

5.4. Extraction of bioactive compounds

AADESs have been also used to extract various food supplements, herbal medicine, and bioactive drug compounds [29,100]. However, some of these show poor bioavailability and solubility in biological systems [101,102]. Duan et al. studied the efficiency of the extraction of some phytochemicals using L-proline-based DESs [29]. Their study indicated that a flavonoid (e.g., icariin) is highly extractable with the L-proline-based DESs compared to sugar or alcohol-based DESs and methanol. Specifically, 1:2 L-proline:levulinic acid, 1:1 L-proline: malonate, and 1:1 L-proline:L-methylurea displayed higher efficiency of icariin extraction. L-proline:levulinic acid AADES showed similar extraction efficiency for anthraquinone to the methanol. On the other hand, L-proline:L-methylurea AADES was found as an excellent solvent to extract the saponins compared to the other amino acid, betaine, and choline chloride-based DESs.

5.5. Pretreatment of the biomass

To enhance the enzymatic hydrolysis of biomass, AADES can play an essential role. Liang et al. (2020) reported the hydrolysis of corncob biomass using 1:1 betaine:amino acid (e.g., arginine, histidine, and lysine) DESs with the water content ranging from 65 to 93% [31]. They pretreated 2 mg of biomass in a covered glass bottle with 1 g of DES by applying 60 °C heat and 350 rpm magnetic stirring. Results showed that lignin and xylan were highly soluble in betaine:lysine and betaine:arginine solvents, and poorly soluble in betaine:histidine DES. Furthermore, betaine:lysine, and betaine:arginine successfully removed about 50% of lignin and xylan after pretreating the biomass at 60 °C for 5 h. The removal of lignin and xylan significantly increased the subsequent efficiency of enzymatic hydrolysis. On the other hand, betaine: histidine removed less lignin and xylan which seems to be insufficient to enhance the enzymatic hydrolysis. In this pretreatment processes, the optimum ratio of solid (biomass) to liquid (DES-water) is another important factor.

5.6. Greener synthesis of bioactive compound

AADES has been used to synthesize biologically active compounds. Indole and its derivatives are present in both marine and terrestrial plants [103]. 3,3-diaryloxindole is one of the indole derivatives, which can be synthesized by the coupling reaction of indole and carbonyl compounds in different solvent media, such as dodecylsulfonic acid [104], silica-sulfuric acids [105], and ionic liquids [106]. All these methods require long time, toxic materials, expensive catalysts, and produce unsatisfactory yields. Therefore, a novel eco-friendly media is needed to synthesize the indole derivative to replace the use of hazardous and toxic chemicals. Research showed that the proline:oxalic acid dihydrate-based DES was used as a greener alternative to synthesize 3,3-diaryloxindole and chromone-based bis(indolyl)alkanes [82]. Excellent yield of the product with high purity was achieved using 1:1 proline:oxalic acid-dihydrate at ambient temperature. They also found that the production of the desired amount of product was reduced at high temperature and in strong acidic medium because indole could be polymerized at those severe conditions. Therefore, optimum reaction condition should be maintained.

6. Summary and future perspectives

Chemical industries are looking for green and sustainable process technologies. One approach is to search for solvents which are alternatives to the inherently toxic and eco-unfriendly organic solvents or the expensive ionic liquids. Therefore, applications of the DESs have been drawn considerable interest from the pharmaceutical industry, engineering, and academia, since DESs are safer, nonvolatile, nonflammable, cheaper to formulate, tunable, typically nontoxic, and a lot of them are applicable to biological system.

Pharmaceuticals primarily rely on oral dosage form, which are mostly solid APIs. Many of the solid APIs have poor solubility, permeability, bioavailability, and thus their therapeutic actions cannot be achieved at the required efficiency in biological systems. These problems could be solved in a less expensive and greener way by formulating them as THEDESs. Depressing the melting point enhances their permeability, intestinal absorption, dissolution rate, and solubility profile. Enhancing their bioavailability could improve the pharmacokinetic and ADMET properties, too. High viscosity of the DESs could be a major hindrance of their wide variety of applications, but addition of glycerol and water or changing the molar ratios of the components could be the easiest ways to overcome the issue. Similar to THEDESs, amino acid-based DESs could also be applicable to improve the bioavailability of various drugs and proteins. Since nonbonding interactions play an important role for the formulation of the DESs, detailed structural insights of the DESs lead to making them more efficient. It is important to know that which compounds have been used for the formulation of the THEDESs and AADESs. Complete knowledge of their formulation, structures, and applications will certainly help to know the gaps in these research areas. Therefore, in this review, we have studied the formulation conditions, molar ratios, compounds used for the formulations, structural features, and the applications of the THEDESs and AADESs. Although both types of DES have mainly been used for the

pharmaceutical applications, AADES have also shown a lot of promise in other fields, such as extraction, synthesis, and separation. Similar to the AADESs, it is also important to explore the application of the THEDESs in the field of synthetic chemistry, catalysis, agriculture, or extraction research.

This review indicates that the advancement and utilization of these emerging DESs could create a safer and greener future for the pharmaceutical, chemical, biochemical, biotechnology, agricultural, and food industries. Some factors, such as addition of excessive water to adjust the viscosity of a DES could eventually lead to reduce its efficacy, their non-bonding interactions could be disrupted in the biological system, and the hygroscopic nature of their components could damage nanostructures and need to be considered during the formulation and the use of the DESs. Therefore, it is essential to conduct extensive studies to optimize and scale-up the reported processes from laboratory to pilot scale. Besides, molecular dynamics (MD) study and multivariate statistical analysis, such as principal component analysis (PCA) and singular value decomposition (SVD) could be applied to make more sense of the existing big data, which will certainly be helpful for the future researchers. It is also important to consider the principles of the green chemistry to make the processes safer and sustainable.

Notes

The authors declare no competing financial interest.

Declaration of Competing Interest

There are no conflicts to declare.

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